

Lab Guide — Lakeflow Connect & Orchestration

Work top to bottom. Everything lives in **your** schema inside the `dev-sh-training` catalog. Set the

`schema` widget to your own name on every notebook before you run it.

- Catalog (fixed): `dev-sh-training`
- Your schema: whatever you set in `00_setup` (e.g. `jane_doe`)
- Landing volume: `/Volumes/dev-sh-training/<you>/landing/`

⚠ **How to use every notebook in this lab — read first.** The widgets (`schema` , `scenario` , ...) are defined in the **first cell**. You must **run that first cell** before anything else so the widget boxes appear at the top of the notebook. Then fill in your `schema` (and any other options) and run the remaining cells (or **Run all**). If you skip the first cell, the widgets never appear, and the notebook falls back to the placeholder default and stops with an error.

The value you type is substituted everywhere automatically — via `${schema}` in SQL notebooks and via the widget in Python notebooks.

Setup (once)

1. Add this repo as a **Git folder** (`Workspace` → `Create` → `Git folder` , URL `https://github.com/realboom/lakeflowconnect`).
 2. Open `notebooks/00_setup` , **run the first cell** so the `schema` widget appears, set it to your name, then **Run all**.
You now have a `schema` and a `landing` volume with `filedrop/` and `sftp/` subfolders.
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Part A — COPY INTO (batch file ingestion)

Concept: `COPY INTO` loads *new* files from a folder into a table and is **idempotent** — rerun it

and already-loaded files are skipped. Simple, batch, great for "a file shows up, pull it in."

1. Open `generate_data`, set `schema`, scenario `Part A - file drop (COPY INTO)`, `num_rows = 25`, **Run all**.
→ writes a comma CSV to `landing/filedrop/incoming/`.
2. Open `part_a_copy_into`, set `schema`, **Run all**.
→ creates `bronze_file_eligibility` and loads the file.
3. **Run the COPY INTO cell again**. It loads **0 new rows** — that's idempotency (it remembers what it already ingested).
4. Generate another Part A file (step 1 again) and re-run `COPY INTO` → only the **new** file is loaded.

✓ Takeaway: `COPY INTO` = the simplest incremental batch ingest.

Part B — Auto Loader as a Job ("SFTP" source), with a retry policy

Concept: A vendor drops files on **SFTP**; we mimic that by landing pipe-delimited files in

`landing/sftp/incoming/` and letting **Auto Loader** (`cloudFiles`) ingest them incrementally (it keeps a

checkpoint of what it has already processed). We run Auto Loader as a **Job with a retry policy** — and

that retry is exactly what lets a schema change self-heal in Part C.

1. **Generate data:** open `generate_data`, scenario `Part B - SFTP source (Auto Loader)`, `num_rows = 25`, **Run all** → writes a pipe-delimited CSV to `landing/sftp/incoming/`.
2. **Create the ingest job:**
3. **Jobs & Pipelines** (left sidebar) → blue **Create** → **Job** → name it `sftp_ingest_<you>`
(jobs are workspace-global, so include your name).
4. Add a task: **Task name** `ingest`, **Type** Notebook, **Source** Workspace, **Path** `notebooks/part_b_autoloader` (Select Notebook → Users → *you* → `lakeflowconnect` → `notebooks`).

5. **Cluster: Serverless** if offered, otherwise **All-Purpose Compute** (keep it warm — you'll run this a few times).
 6. **Parameters:** + Add → Key `schema` , Value = your schema.
 7. **Retries:** click the  edit → set **Retry at most = 1, and wait 30 seconds** (change the unit from `mins` to `secs`). On serverless, also **uncheck "Enable serverless auto-optimization"** so your one retry is the only retrier → **Confirm**. This single retry is what makes the schema change in Part C self-heal (and stay a clean *one* retry).
 8. **Create task**, then **Run now**.
 9. Open the **Runs** tab → the run succeeds and `bronze_sftp_eligibility` is loaded.
 10. Generate another Part B file (step 1) and **Run now** again → only the **new** file is picked up (checkpoint).
- ✓ Takeaway: Auto Loader = incremental, checkpointed ingestion for flat-file / "SFTP" feeds. Contrast with Part A: COPY INTO is batch SQL; Auto Loader is a streaming source with schema evolution + rescue (next) — and running it as a **Job** gives you retries (Part C), scheduling, and triggers (Part E).
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Part C — Auto Loader resilience (run via your `sftp_ingest` job)

Same `bronze_sftp_eligibility` . For each scenario, generate the file, then **Run now** your `sftp_ingest_<you>` job and watch the **Runs** tab. **Rescued data first** (original schema), then schema evolution (which permanently adds a column — and shows your retry policy in action).

C1 — Rescued data (bad value, original schema)

1. `generate_data` → scenario `Part C - rescued data (SFTP)` → **Run all**. Half the rows have a bad `eff_date` of `NOT-A-DATE` . (`eff_date` is typed `DATE` via `schemaHints` .)
2. **Run now** your `sftp_ingest_<you>` job. The bad values can't cast to DATE, so Auto Loader **rescues** them: the row still ingests with `eff_date = NULL` , original value parked in `_rescued_data` . One clean attempt — no retry needed (no schema change).

3. **See it without running anything:** open the run → click the `ingest` task → scroll the notebook output. The ingest notebook already displays the **rescued rows** (`_rescued_data` populated, `eff_date` NULL) in its last cell. (No separate query needed.)

💡 A malformed value didn't crash the pipeline or get silently dropped — it's captured for audit/backfill.

C2 — Schema evolution (a new column) — watch the retry heal it

1. `generate_data` → scenario `Part C - schema evolution (SFTP)` → **Run all**. The file has a **new** `risk_tier` column.
2. **Run now** your `sftp_ingest_<you>` job and watch the **Runs** tab. Auto Loader detects `risk_tier`:
Attempt 1 fails (Auto Loader stops to *record* the new column), then the **retry** — **Attempt 2** — **succeeds**, with `risk_tier` added to the table (existing rows `NULL`). The fail-then-recover is visible right in the run.
3. **See the new column:** click into the run → the `ingest` task → the **Attempt 2 (succeeded)** notebook output shows the data **with the new** `risk_tier` **column** populated. (Attempt 1 is right there too, showing the schema-change failure — nice to point at.)

💡 A new field appeared and the table evolved on its own — the task **retry** turned a one-time schema handshake into a hands-off success. (This is the Part E6 retries lesson, shown live.)

⚠️ **If the run just succeeds with no failed attempt**, `risk_tier` was already in Auto Loader's schema from a previous run — so there's no "new column" to record and nothing to retry. The fail→retry only fires when `risk_tier` is **new**. To replay cleanly: run `reset` (your schema) — it clears the landing files **and** the `_autoloader` checkpoint/schema **and** drops `bronze_sftp_eligibility` — then run **Part B** (re-establishes the schema *without* `risk_tier`), then C2. (Clearing just the landing files isn't enough; the evolved schema lives in the checkpoint.)

Part E — Orchestration (Lakeflow Pipelines + Jobs)

Now we orchestrate the eligibility data you just created. You'll build **one pipeline** and a few **jobs** by hand.

Where you build these — read this once. In the Databricks **left sidebar**, find **Jobs & Pipelines**.

Tip: **right-click it and open in a new browser tab**, so you can keep this guide open side-by-side.

On the **Jobs & Pipelines** page, click the blue **Create** button (it has a dropdown). The dropdown lets

you pick what to build — **ETL Pipeline** or **Job**. Each step below tells you which one to choose:

- **E1** builds an **ETL Pipeline** (that's what a Spark Declarative Pipeline is).
- **E2, E3, E5** build **Jobs**.

Picking a notebook for a task — read this once too. Whenever a task asks for a notebook **Path**, the

Select Notebook dialog opens. Navigate **Workspace** → **Users** → *your user* →

`lakeflowconnect` →

`notebooks`, select the file, and click **Confirm**. (`lakeflowconnect` is the Git folder you cloned in

Setup.) Below, a reference like `notebooks/gold_eligibility_summary` means that file in this folder.

E1 — The silver pipeline (an ETL Pipeline / Spark Declarative Pipeline)

This builds a streaming `silver_eligibility` table from your Auto Loader bronze (`bronze_sftp_eligibility`).

1. **Jobs & Pipelines** (left sidebar) → blue **Create** button → **ETL Pipeline**. You land in the pipeline editor on a new pipeline named like *New Pipeline 2026-...*, with a starter file `transformations/my_transformation.py`.
2. **Rename the pipeline:** click its name at the top-left and change it to `silver_eligibility_<you>`.
If it asks "**Also rename the root folder?**", click **Rename** (keeps the pipeline's workspace folder name in sync — either choice works).
3. **⚠ Set the default catalog + schema — this is how the pipeline finds your tables.**
Open the pipeline **Settings** (gear icon, or the `:` menu → *Settings*) and set:
4. **Default catalog =** `dev-sh-training`

5. **Default schema = your schema** (the one from `00_setup`)

6. **Serverless** compute

(The editor shows a default like `... / default` near the top-right — you must change it to yours.)

Because the code uses the bare name `bronze_sftp_eligibility` (no catalog/schema), it resolves

against this default catalog + schema — so your pipeline reads *your* bronze and writes *your*

silver. Same code for everyone; only the default schema differs.

4. **Add the transformation SQL.** The starter file is Python, but our logic is SQL:

– Right-click `my_transformation.py` → **Rename** to `silver_eligibility.sql` (or click **+** to add

a new `silver_eligibility.sql` and delete the empty `.py`).

– Open `notebooks/silver_eligibility` in your Git folder, copy the

`CREATE OR REFRESH STREAMING TABLE silver_eligibility ...` statement, and paste it into the file.

– (Optional — AI authoring: instead of pasting, click *Create with Genie Code / Generate and*

paste this prompt — then compare what it writes to the SQL in the repo before running:

```
> "Create a streaming table in SQL named silver_eligibility that reads incrementally from the  
> streaming table bronze_sftp_eligibility. initcap first_name and last_name; cast eff_date and  
> term_date to DATE; keep member_id, plan_code, line_of_business; add silver_loaded_at as  
> the current timestamp. Use FROM STREAM bronze_sftp_eligibility so it's incremental, not a reload."  
)
```

5. Click **Run** (▶ Run pipeline) → it builds `silver_eligibility` in your schema from `bronze_sftp_eligibility`. (Part B must have run first so your bronze table exists.)

E2 — Table-update trigger (event-driven)

Make the silver pipeline run **automatically whenever new bronze data lands** — no schedule, no manual run.

1. **Jobs & Pipelines** (left sidebar) → blue **Create** button → **Job**. You land in the job editor on a job named like *New Job 2026-...*

2. **Rename the job:** click its name at the top-left → `silver_refresh_<you>` .

3. **Add the task — and make it a Pipeline task** (the screen defaults to a Notebook card; don't use that):

4. Click **+ Add another task type** (the blue button under "Add your first task").
 5. **Task name:** `refresh_silver`
 6. **Type:** choose **Pipeline** from the Type dropdown.
 7. **Pipeline:** select your `silver_eligibility_<you>` pipeline from E1.
 8. **Create task / Save task.**
 9. **Add the trigger** — it lives in the **right-hand panel**, not on the task:
 10. In the right-side **Job details** panel, find **Schedules & Triggers** and click **Add trigger**.
 11. **Trigger type: Table update.**
 12. **Tables:** click the table box and browse to `dev-sh-training` → **your schema** → `bronze_sftp_eligibility`. ⚠️ Make sure it's `bronze_sftp_eligibility` (the Auto Loader table your silver pipeline streams from) — **not** `bronze_file_eligibility` (that's Part A's COPY INTO table; silver doesn't read it, so the job would fire but silver would see nothing new).
 13. **Advanced** is optional. The **Minimum time between triggers / Wait after last change** fields are in `hh mm` (not seconds) — leave both at `00h 00m` so the job fires as soon as new data lands.
 14. **Save.**
 15. **Test it:**
 16. Click the **Runs** tab (next to **Tasks**, just under the job name) so you can watch runs appear.
 17. In another browser tab, run `generate_data` (scenario **Part B – SFTP source**) then `part_b_autoloader` so new rows commit to `bronze_sftp_eligibility`. (Plain Part B is all you need — you just want a fresh commit to fire the trigger. `risk_tier` already exists from Part C and silver doesn't use it, so there's no need to re-run schema evolution here.)
 18. Within ~a minute a run **shows up on the Runs tab on its own** — its origin reads *"Triggered by table update"* — and it refreshes `silver_eligibility`. (No one pressed Run now.)
- 💡 The trigger baselines when you create it and fires only on **new** commits — so do the fresh bronze load *after* you've set up the trigger, or it won't have anything new to react to.

E3 — The DAG: gold → report → notify (with a conditional branch)

Create a multi-task job.

1. **Jobs & Pipelines** → blue **Create** button → **Job**, name it `eligibility_gold_<you>` (include your name — jobs are workspace-global, so a bare `eligibility_gold` would collide with everyone else's).

Task fields — set these, ignore the rest. Each task's config panel has a lot of fields; you only touch a few:

- **Task name**, **Type** = Notebook, **Source** = Workspace, **Path** (via the Select Notebook dialog).
- **Cluster** → open the dropdown. If **Serverless** is offered, pick it. **Jobs serverless may not be enabled in this workspace** — if there's no Serverless option, choose the existing **All-Purpose Compute** cluster (already running = fastest), or **Add new job cluster** if no shared cluster is available (works, but ~5-min cold start). **Use the same cluster for every task in the job.**
- **Parameters** → click **+ Add** and enter the Key/Value shown in each step (this is how the notebook gets your `schema`).
- **Retries** → only where a step says so.

Leave **Dependent libraries**, **Notifications**, **Metric thresholds**, **Tags**, **Job health**, **Permissions**, and **Advanced settings** at their defaults.

1. Task `gold` — Type **Notebook**, Path `notebooks/gold_eligibility_summary` , **Cluster** = your compute (see note). **Parameters:** **+ Add** → Key `schema` , Value = your schema. **Retries:** **+ Add** → **Retries:** edit → **Retry at most = 2**, wait 30 seconds.
2. Task `report` — **+ Add task** , Type **Notebook**, Path `notebooks/eligibility_report` , same cluster, Parameter `schema` = your schema. **Depends on:** `gold` .
3. Task `notify_pass` — Type **Notebook**, Path `notebooks/notify` , same cluster, Parameter `result` = `pass` . **Depends on:** `report` . **Run if = All succeeded.**
4. Task `notify_fail` — Type **Notebook**, Path `notebooks/notify` , same cluster, Parameter `result` = `fail` . **Depends on:** `report` . **Run if = At least one failed.**

5. **Run** the job → gold → report → notify_pass (and notify_fail is skipped).
Green path.

E4 — Repair / partial re-run (recover from a schema change)

A realistic failure: someone renames a column upstream and the downstream report breaks.

1. Open `gold_eligibility_summary` and rename `member_count` → `enrolled_members` in **both cells** —

the `count(*) AS member_count` in the `CREATE OR REPLACE TABLE` cell **and** the `ORDER BY member_count`

in the final check cell. Save. (If you miss the check cell, the `gold` task itself errors there and you

won't get the "gold succeeds / report fails" story.)

2. **Run** `eligibility_gold_<you>` . → `gold` **succeeds** (table rebuilt with the new name), but `report`

fails: `UNRESOLVED_COLUMN ... member_count ... Did you mean 'enrolled_members'?`, and `notify_fail` runs.

3. **Fix** `eligibility_report` : change `member_count` → `enrolled_members` , save.

4. On the failed run click **Repair run.** → `gold` **is skipped** (already built), only `report` + `notify_pass` re-run → success.

💡 You don't rerun the expensive upstream that already succeeded — fix and resume from the failure.

(To reset for a repeat: revert both files back to `member_count` .)

E5 — For each (fan-out + concurrency)

Run a task once per line of business, in parallel. **Read this once:** a **For each** task is a *wrapper* —

you set the list to iterate over on the wrapper, then add the task that actually runs **inside** the loop.

It's two levels, and the second one is easy to miss.

1. **Jobs & Pipelines** (left sidebar) → blue **Create** button → **Job**. You land in the job editor on a job named like *New Job 2026-....*
2. **Rename the job:** click its name at the top-left → `eligibility_by_lob_<you>` .
3. **Add the For each wrapper task.** Click **+ Add another task type** (the blue button under "Add your first task"), then set **Type** to **For each**. Fill the form:
4. **Task name:** `for_each_lob`
5. **Type:** **For each** (already selected)

6. **Inputs:** replace the `[1,2,3]` placeholder with `["Commercial","Medicaid","Medicare Advantage","Individual"]`
7. **Concurrency:** `4`
8. Leave **Notifications** empty.
9. **⚠ Add the task that runs *inside* the loop — this is the step people miss.** The For each you just filled is only the wrapper; the real work goes in the dashed **"For each loop"** box on the canvas above the form. Click the `+` *inside that dashed box* (**not** the job-level "Add task"). A new task card opens:
10. **Task name:** `build_gold_lob`
11. **Type:** **Notebook**
12. **Path:** `notebooks/for_each_lob` (use the **Select Notebook** navigation from the top of Part E).
13. **Parameters** → click **Add** and create **two** key/value rows:
 - `schema` = **your schema** (the one from `00_setup`)
 - `lob` = `{{input}}` — the per-iteration value; the loop passes one element of **Inputs** into each run. Type the double curly braces exactly (`{{input}}`), or every iteration gets the literal text instead of a line of business.
14. **Save task.**
15. **Run now** → four iterations fan out **in parallel** (concurrency `4`), each building `gold_elig_<lob>`
 — `gold_elig_commercial` , `gold_elig_medicaid` , `gold_elig_medicare_advantage` , `gold_elig_individual` —
 filtered to that line of business.
16. **See the four iterations (you have to drill in — the Runs list won't show them).**
 Open the **Runs** tab → **click the run that's currently executing** (the active row at the top). That opens the run graph;
 now **click the** `for_each_lob` **task** → the right-hand panel switches to the **For each / iterations view**, showing all four iterations (one per LOB) with their individual status, each running `build_gold_lob` . (At the top Runs level you only see the single parent job run — the per-LOB breakout lives one level down, inside the For each task.)

💡 One task definition, four parallel runs — fan-out keyed off a simple array, each run isolated to its own LOB.

Concurrency FAQ (you'll be asked): concurrency is **1–100** (default 1); effective parallelism =

`min(concurrency, # inputs)`; the workspace cap is **2,000 concurrent task runs**.

Size it to what your

downstream can handle, not the max. See the instructor runbook for the full FAQ.

E6 — Retries

You already saw retries do real work back in **Part C2**: when the `risk_tier` column appeared, your

`sftp_ingest` job's **Attempt 1 failed** (Auto Loader stopping to record the new column) and the **retry**,

Attempt 2, succeeded — the schema evolved with no manual intervention. That's the **Retry at most = 1**

you set on the ingest task (and **Retry at most = 2** on `gold` in E3). Retries also heal transient infra

failures. Put a retry on any task that ingests evolving files or calls flaky external systems — then open a

run and show the **Attempt 1 → Attempt 2** timeline.

E7 — Incremental loading: upsert with AUTO CDC (*optional — a preview of the SDP workshop*)

The E1 `silver_eligibility` table is **append-only**: re-send a member and you get a **duplicate** row.

A "current state" table (eligibility, customer, account) usually wants **one row per member, updated in**

place. Databricks does this with **AUTO CDC** (the SQL `APPLY CHANGES` pattern) — you declare the

key and a **sequence**, and Databricks writes the insert-or-update MERGE for you (no hand-coded MERGE, no full reload).

1. Build a **second ETL Pipeline**, set up exactly like E1 (Create → ETL Pipeline → rename, e.g.

`silver_upsert_<you>` → **default catalog** `dev-sh-training` + **your schema** → Serverless), but paste

`notebooks/silver_eligibility_scd` as the source. It creates

`silver_eligibility_current`

(one upserted row per `member_id`). Click **Run**.

2. Note a few members and their plans:

```
SELECT member_id, plan_code FROM silver_eligibility_current ORDER BY  
member_id LIMIT 5;
```

3. **Send updates:** generate_data → scenario Part E - member update (upsert) → Run all. It

re-sends existing members with a **new plan_code**. Then run **part_b_autoloader** so the changes land in bronze.

4. **Run the AUTO CDC pipeline again** (or wire a table-update trigger like E2 to fire it).

Re-check the same members — their **plan_code** has **changed in place**: same row count, **no duplicates**.

5. **See the contrast:** the append **silver_eligibility** from E1, fed the same update, shows **two rows**

for that member (old + new). Append vs. upsert is the core incremental-loading choice.

6. **Validate what each run actually changed (CDF).** The pipeline only shows row counts, not rows — so

a successful upsert can look like "nothing happened." To see the inserts vs. updates per run, run

notebooks/silver_eligibility_current_audit (set your **schema**), or run this directly against your schema:

```
sql -- inserts vs. updates per run (commit version ≈ one pipeline run) --  
_change_type: insert | update_preimage | update_postimage | delete SELECT  
_change_type, _commit_version, COUNT(*) AS record_count FROM  
table_changes('silver_eligibility_current', 1) WHERE _change_type <>  
'update_preimage' -- drop the "before" image so each change counts once  
GROUP BY _change_type, _commit_version ORDER BY _commit_version DESC,  
_change_type;
```

The first load is all **insert**s; after the member-update run (steps 3–4) you'll see **update_postimage**

rows for the changed members on a **new version**.

⚠ **CDF has to be on the table.** **notebooks/silver_eligibility_scd** already sets

TBLPROPERTIES ('delta.enableChangeDataFeed' = 'true') on **silver_eligibility_current** — but if you built the **silver_upsert_<you>** pipeline *before* that line existed, **full refresh** the pipeline once so the property takes effect. (If **table_changes(..., 1)** errors that CDF

wasn't enabled at
version 0, start from a later version or a timestamp '...' instead.)

The three pieces to call out: `KEYS (member_id)` (match on the business key),
`SEQUENCE BY ingested_at`
(newest wins when a key repeats), `AUTO CDC INTO` (Databricks runs the MERGE). This
is the foundation we go
deep on in the SDP workshop — upserts, deletes (`APPLY AS DELETE`), and SCD Type 2
history.

Clean up / start over

Run `notebooks/reset` (set your `schema`) to drop the lab tables and clear the landing
folders.

Your schema and volume stay. Re-run `generate_data` to begin again.